Protein Engineering

Module: SC3282

By

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Lecture 01

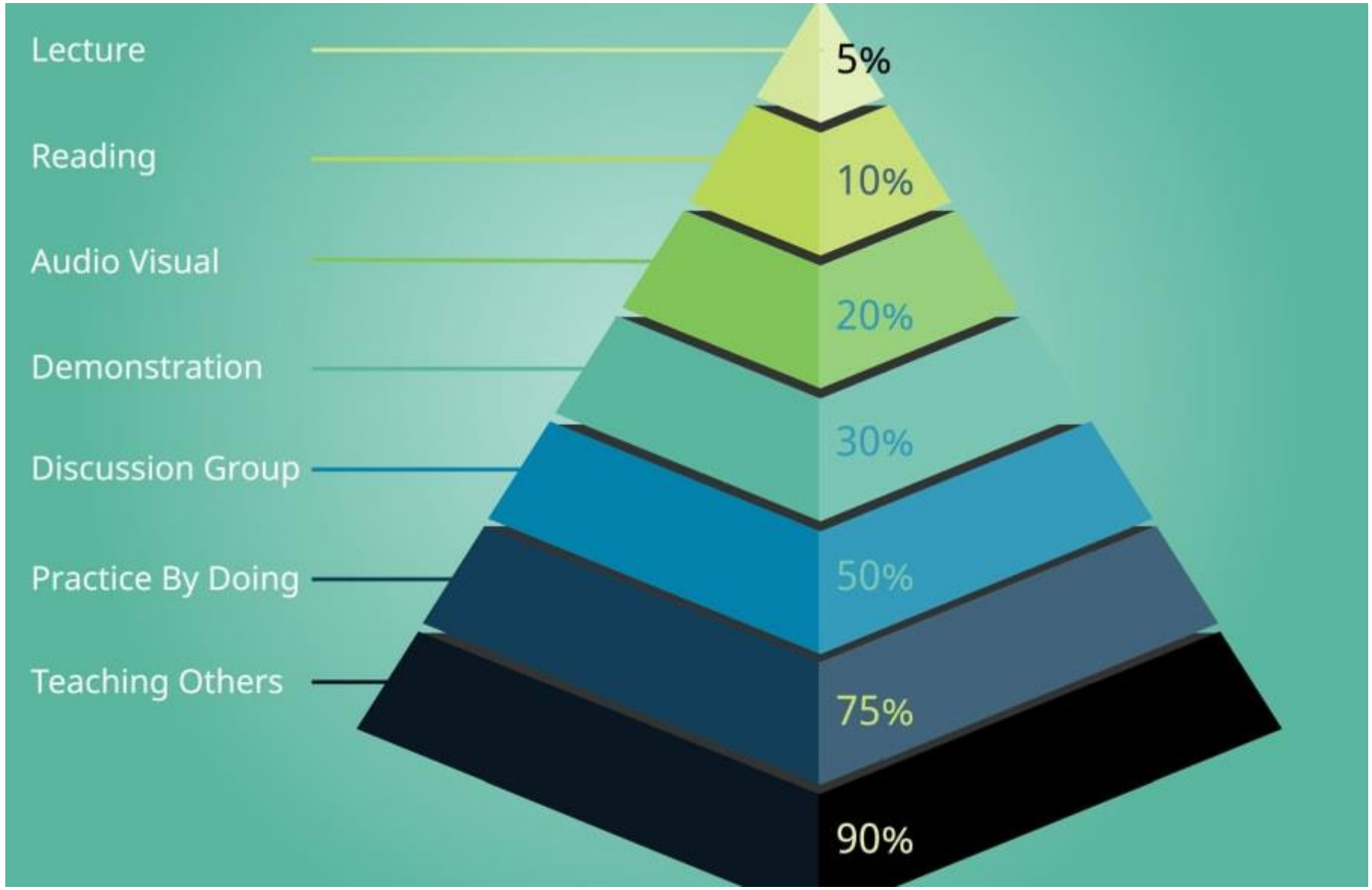
Introduction to protein engineering

Learning outcome of this lecture

- Understanding basic structure of the proteins
- Objectives of the protein engineering
- Application in protein engineering

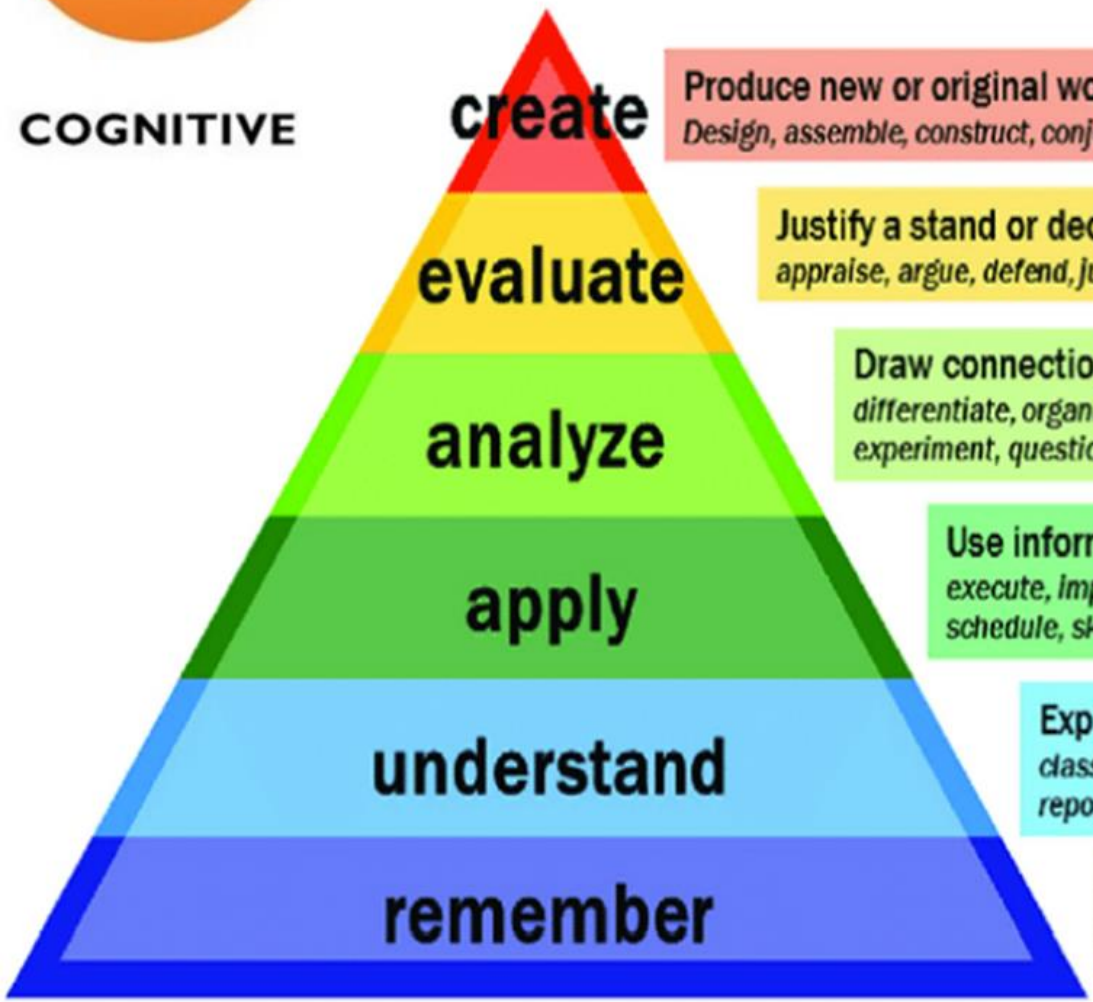
Assignments

1	Protein engineering in food industry
2	Protein engineering in medicine
3	Protein sequencing methods
4	Antibody engineering
5	Protein quantification methods
6	Protein expression systems
7	Protein Purification Methods
8	Protein structure determination


Lecture
5%
Reading
10%
Audio Visual
20%
Demonstration
30%
Discussion Group
50%
Practice By Doing
75%
Teaching Others
90%

Bloom's Taxonomy

COGNITIVE



Produce new or original work
Design, assemble, construct, conjecture, develop, formulate, author, investigate

Justify a stand or decision
appraise, argue, defend, judge, select, support, value, critique, weigh

Draw connections among ideas
differentiate, organize, relate, compare, contrast, distinguish, examine, experiment, question, test

Use information in new situations
execute, implement, solve, use, demonstrate, interpret, operate, schedule, sketch

Explain ideas or concepts
classify, describe, discuss, explain, identify, locate, recognize, report, select, translate

Recall facts and basic concepts
define, duplicate, list, memorize, repeat, state

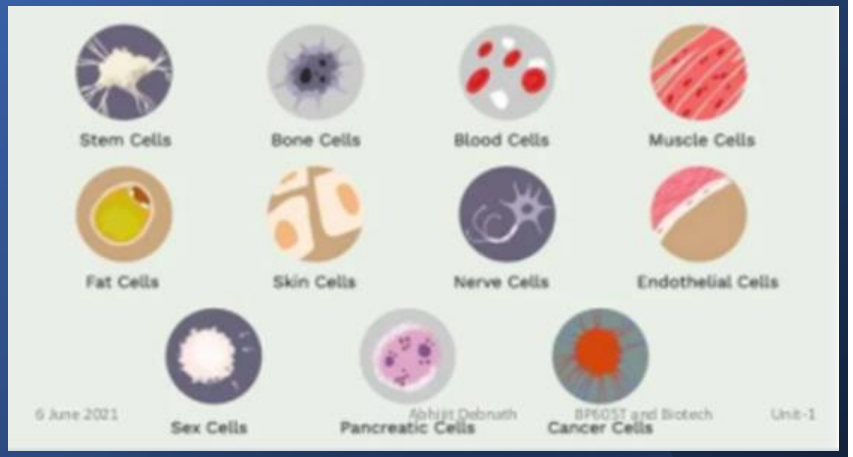
Module information

Module Title	Protein Engineering				
Module code	SC3282	Total credits	02	Tutorial hours	05
Pre-requisite Modules	Food Biotechnology	Lecture hours	30	Independent Learning hours	45
		Practical hours	00		
Module type	Core	Assignment hours	20	Total hours	100

How we are going to assess you?

Methods of Assessment	Description	No. Assessments / Semester	Weightage%
Continuous			**40**
Assignments/tutorial	Assignment	1	10
Quizzes	2 quizzes	2	10
Mid Semester			20
Final Examination			6**0**

Introduction to protein engineering



What is a protein?

- Proteins perform a vast array of functions within organisms, including catalysing metabolic reactions,
  - DNA replication,
  - responding to stimuli,
  - providing structure to cells and organisms, and
  - transporting molecules from one location to another.
- Proteins are large biomolecules and macromolecules that comprise one or more long chains of amino acid residues.
- Proteins are differ from one another primarily in their sequence of amino acids.

Functions of proteins

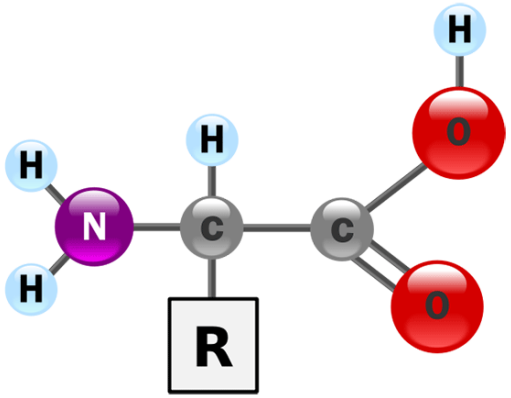
- Enzymes- Metabolism
- Structural- Collagen and Keratin
- Cell Recognition- proteins on cellular surface.
- Regulation of Gene Expression- Gene Repressors or Enhancers.
- Defense- Antibodies.

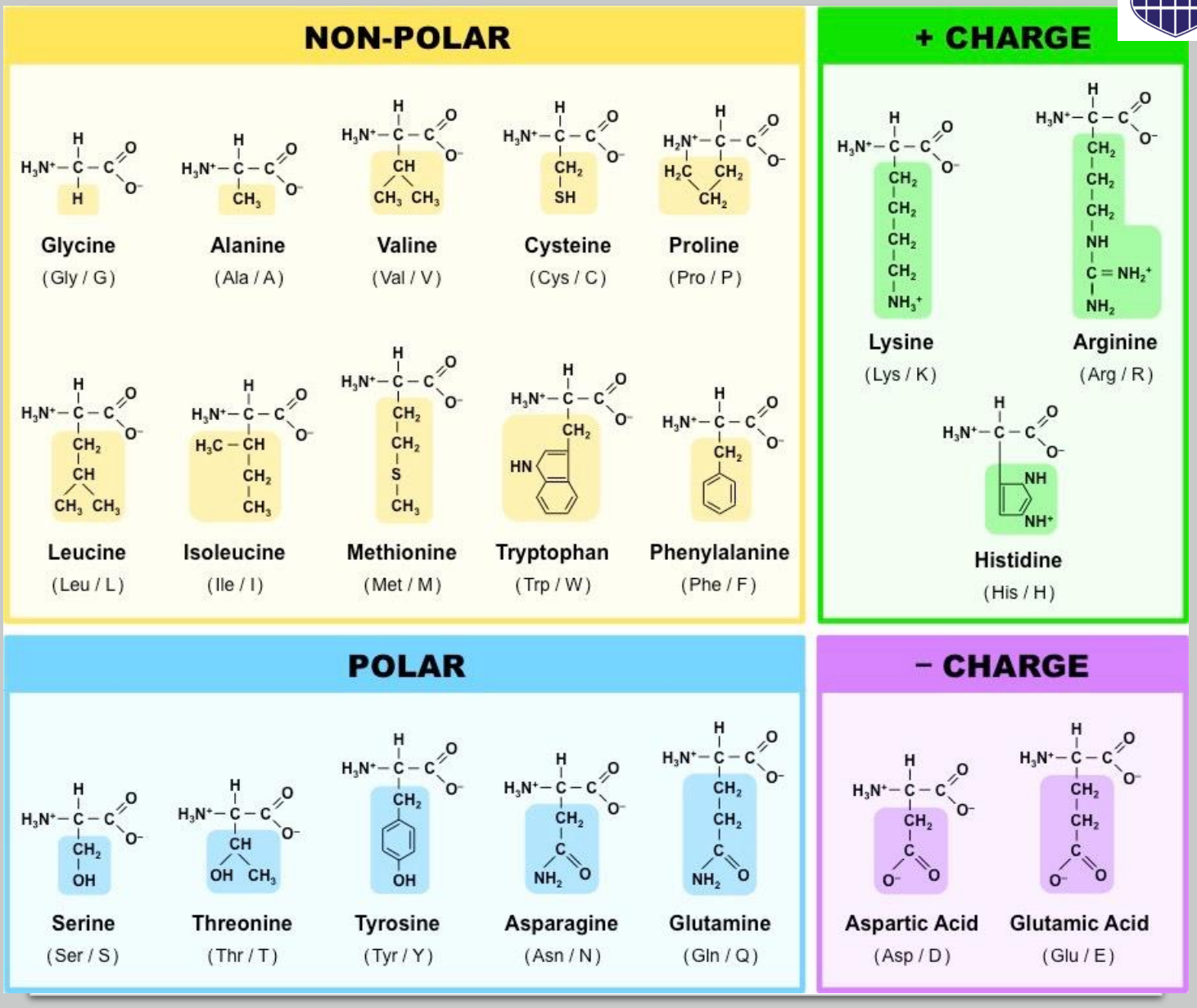
A gross molecular contents of a human cell

Molecule	Percent of Mass
DNA	0.1
Other Organics	0.4
RNA	1.0
Other Inorganics	1.5
Lipids	12
Protein	20
Water	65

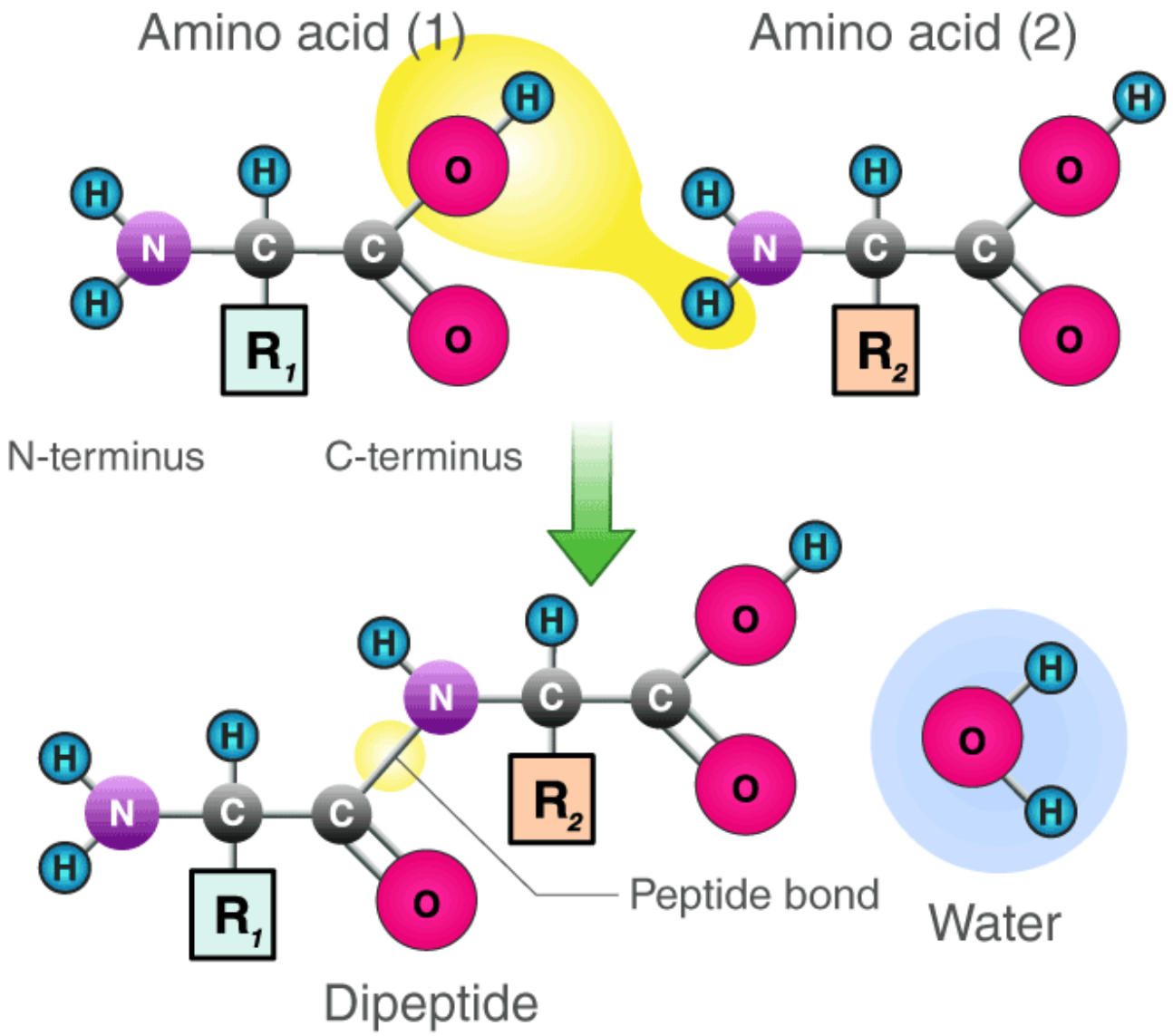
Proteins are made up from amino acids

- Peptide has 50 or less amino acids.
- Protein has more than 50 amino acids.

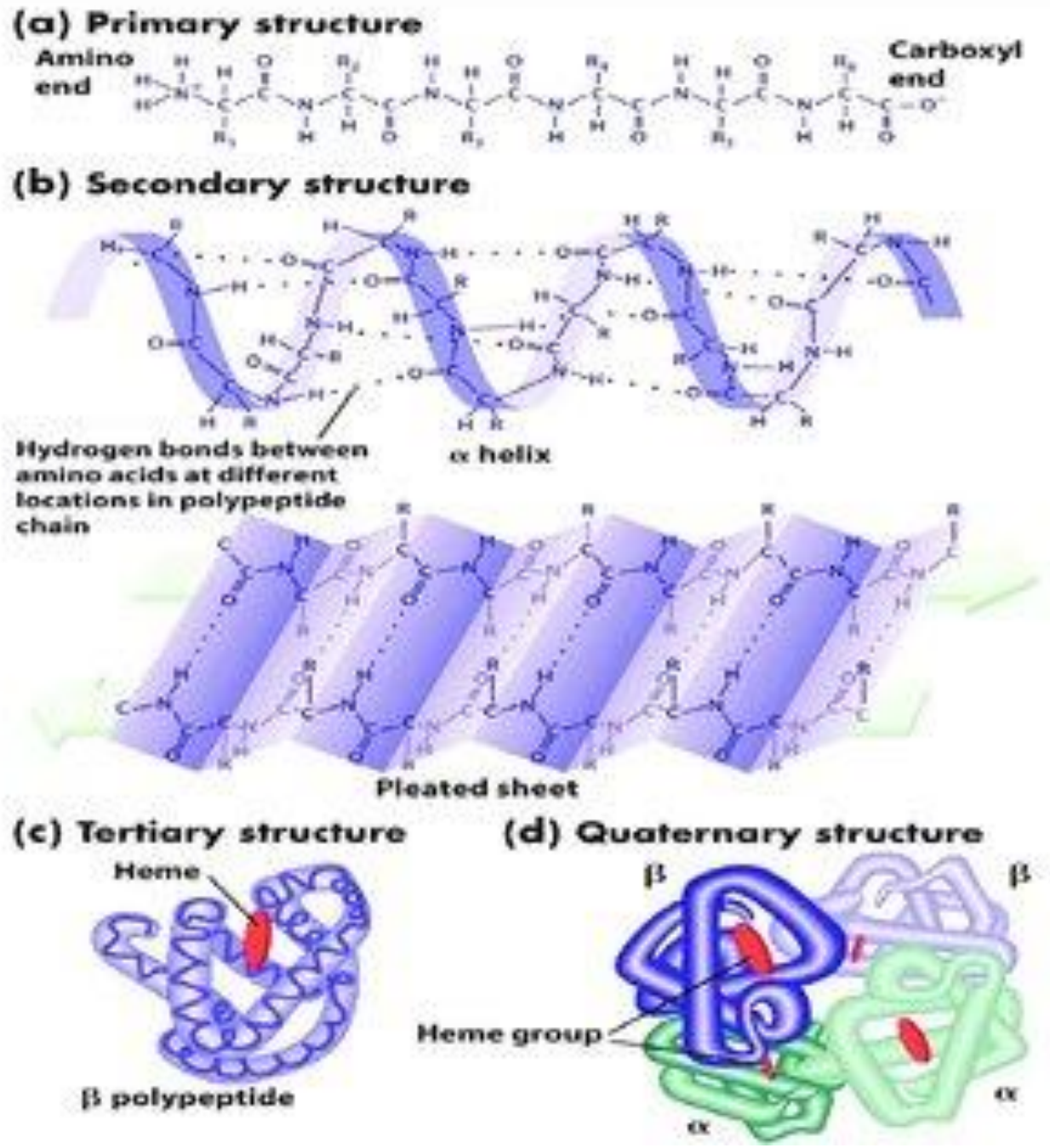




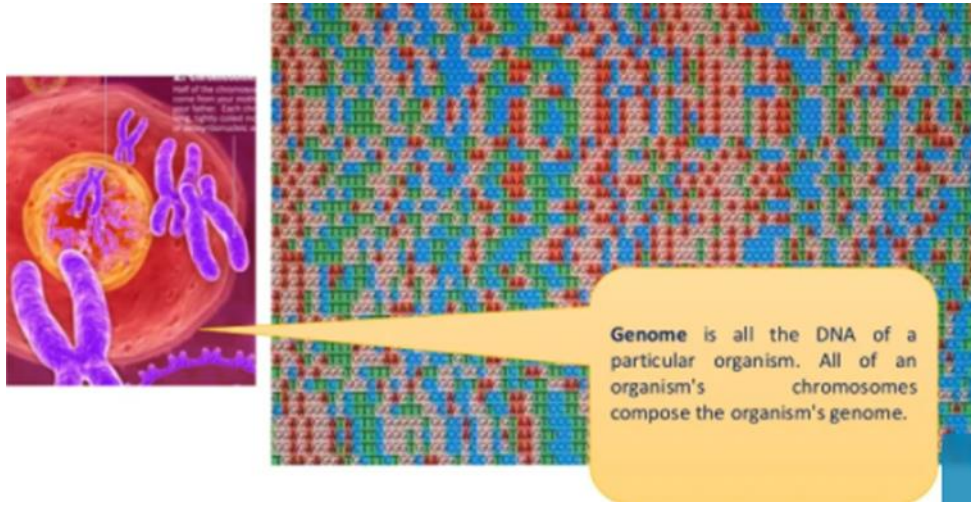
Peptide bond

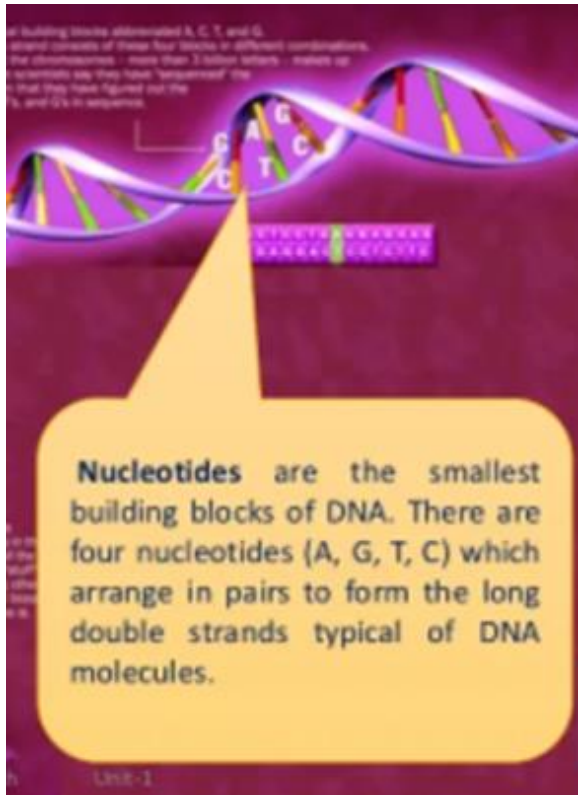


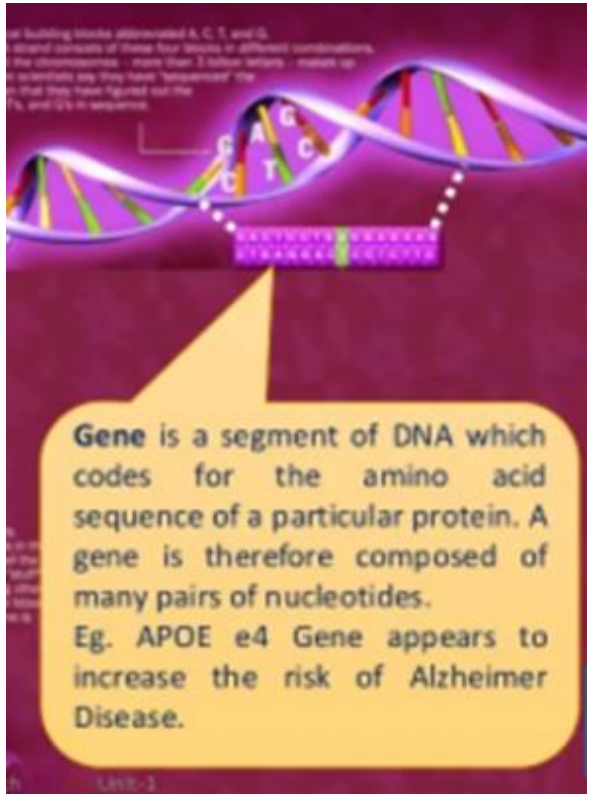
Protein structures

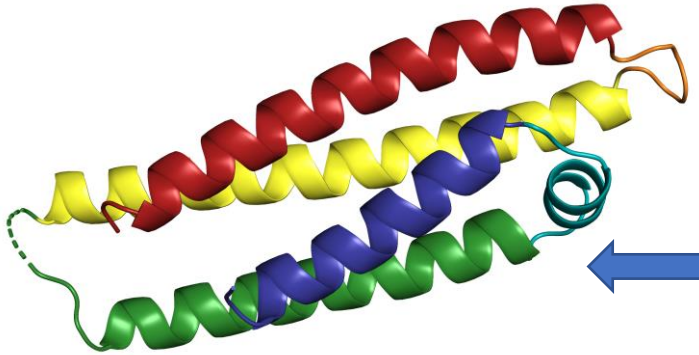


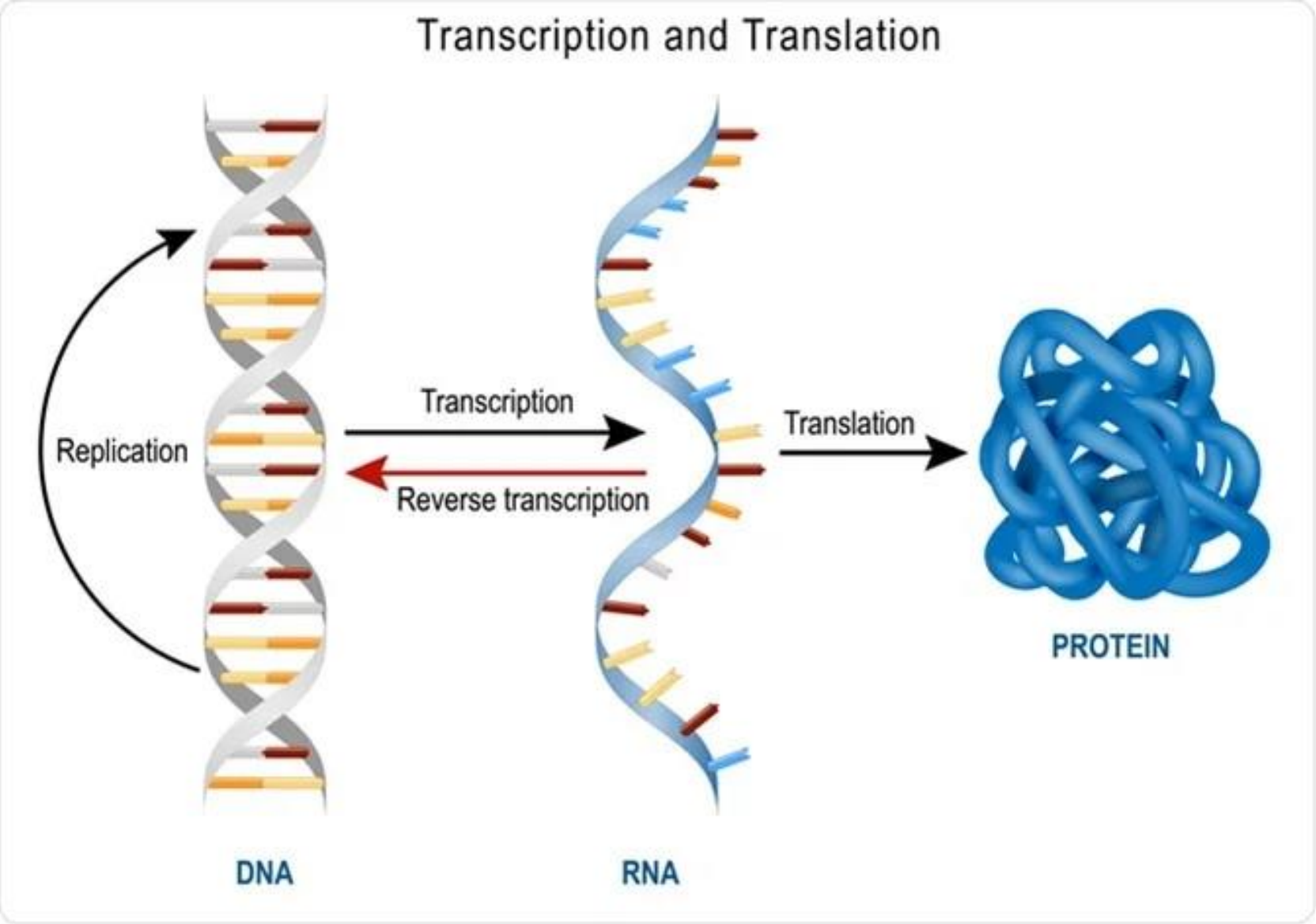
From genome to protein



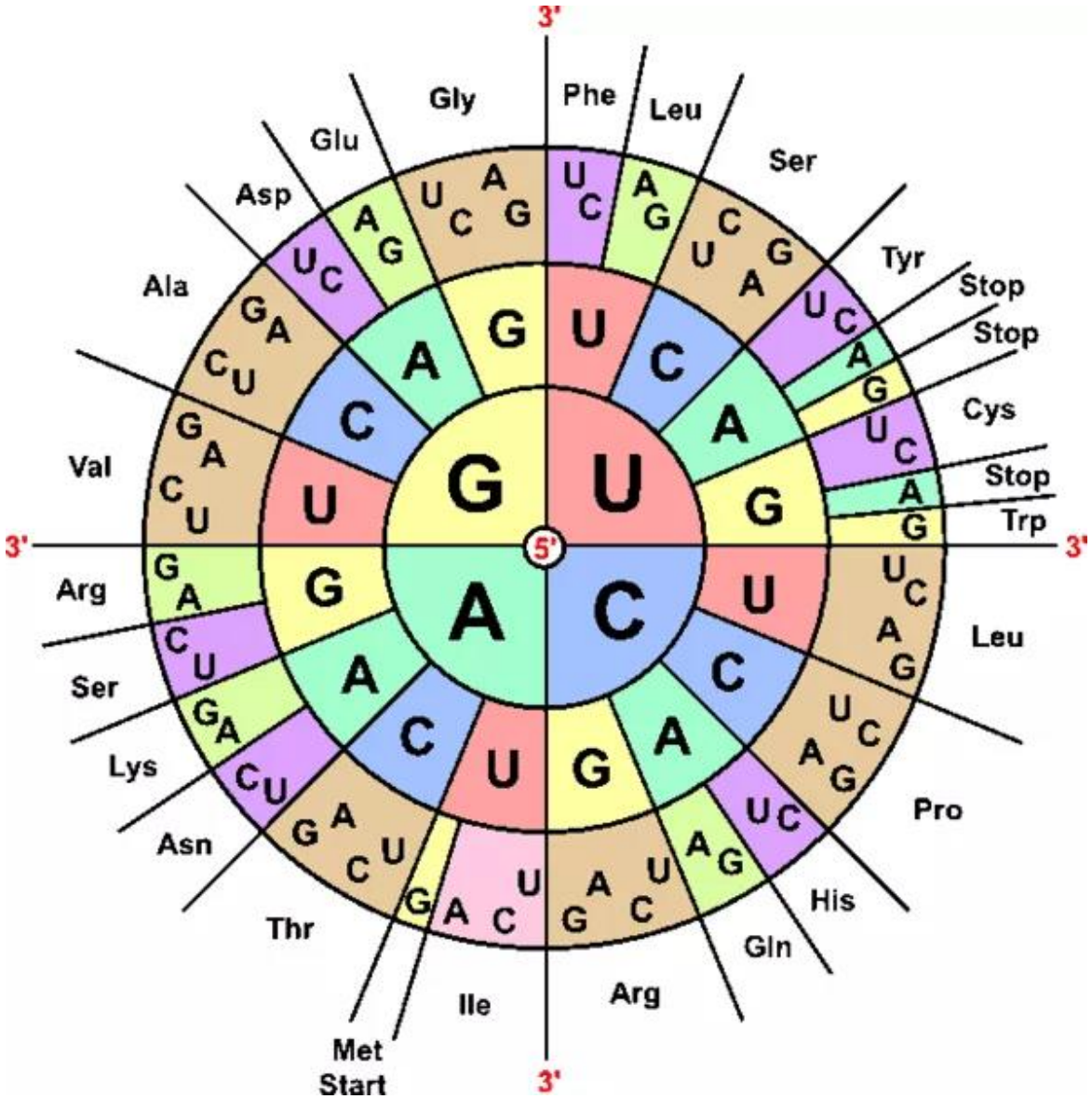







Transcription and Translation
Replication
Transcription
Reverse transcription
Translation
PROTEIN
DNA
RNA

RNA codon amino acid chart.

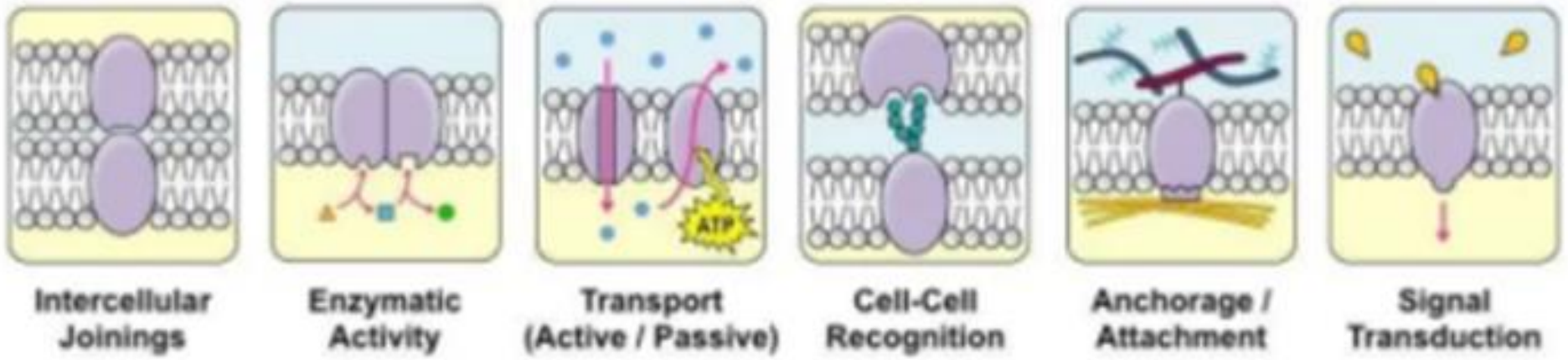


- In biology, a codon refers to the trinucleotides that specify for a particular amino acid.
- For example, Guanine-Cytosine-Cytosine (GCC) codes for the amino acid alanine.
- The Guanine-Uracil-Uracil (GUU) codes for valine.
- Uracil-Adenine-Adenine (UAA) is a stop codon.
- The codon of the mRNA complements the trinucleotide (called anticodon) in the tRNA.

Facts:

- There are,
  - 37.2 trillion cells in Human Body.
  - 23 chromosome pairs.
  - 3 billion DNA base pairs.
  - About 25,000 genes are present in Human Genome
  - About 20,000 of these genes are protein-coding genes.
  - Human make at least 20,000 proteins.
  - 21 Amino acids are the building blocks of the proteins.

Membrane protein functions



What is protein engineering?

- Protein engineering involves synthesis of new proteins from existing protein.
- Novel protein design algorithms, advancement in structural bioinformatics, use computational approaches for protein engineering.
- It is a young discipline, It is also a product and services market, with an estimated value of $168 billion by 2017.
- Protein engineering modifies a protein sequence through substitution, insertion, or deletion of nucleotides, modified protein that is more suitable for a particular application or purpose than the unmodified protein.

Why we need protein engineering

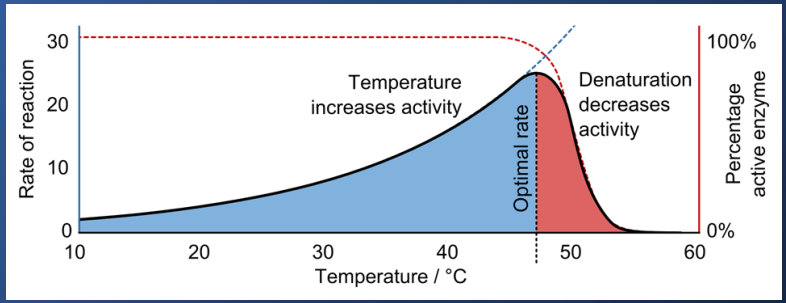
- It is useful for studies of basic research and application, in structure-function relations and for exploitation in industry.
- Genetic engineering provides proteins in bulk which are processed and redesigned to make them suitable for pharmaceutical and industrial requirements.
- Protein engineering can be defined as the modification of protein structure with recombinant DNA technology or chemical treatment to get a desirable function for better use in medicine, industry and agriculture.

The objectives of protein engineering

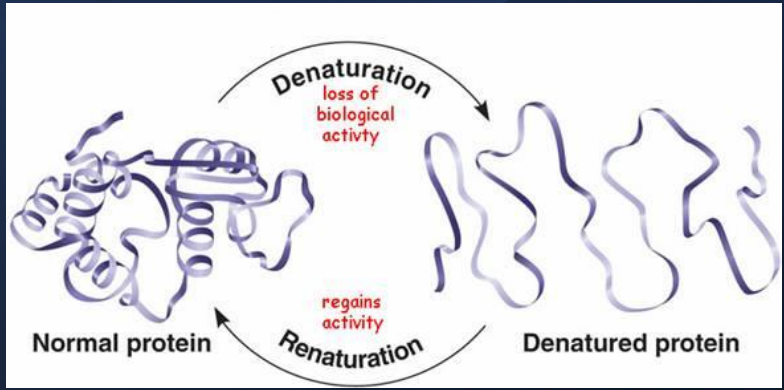
To create a superior enzyme to catalyze the production of high value specific chemicals.

To produce enzyme in large quantities.

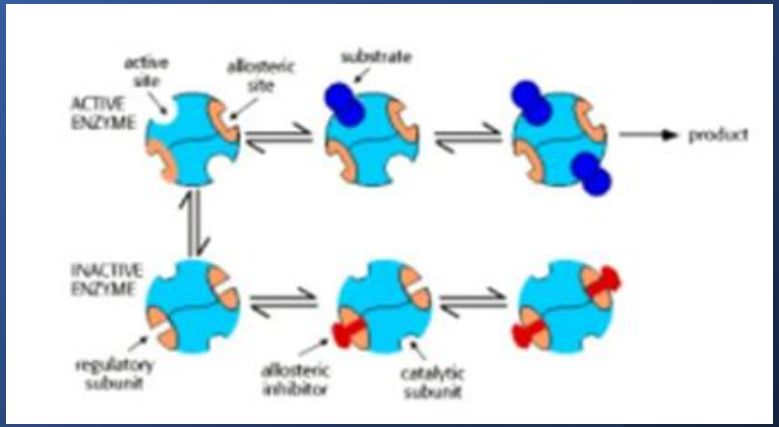
To produce biological compounds (include synthetic peptide, storage protein, and synthetic drugs) superior to natural one.



Rationale of protein engineering



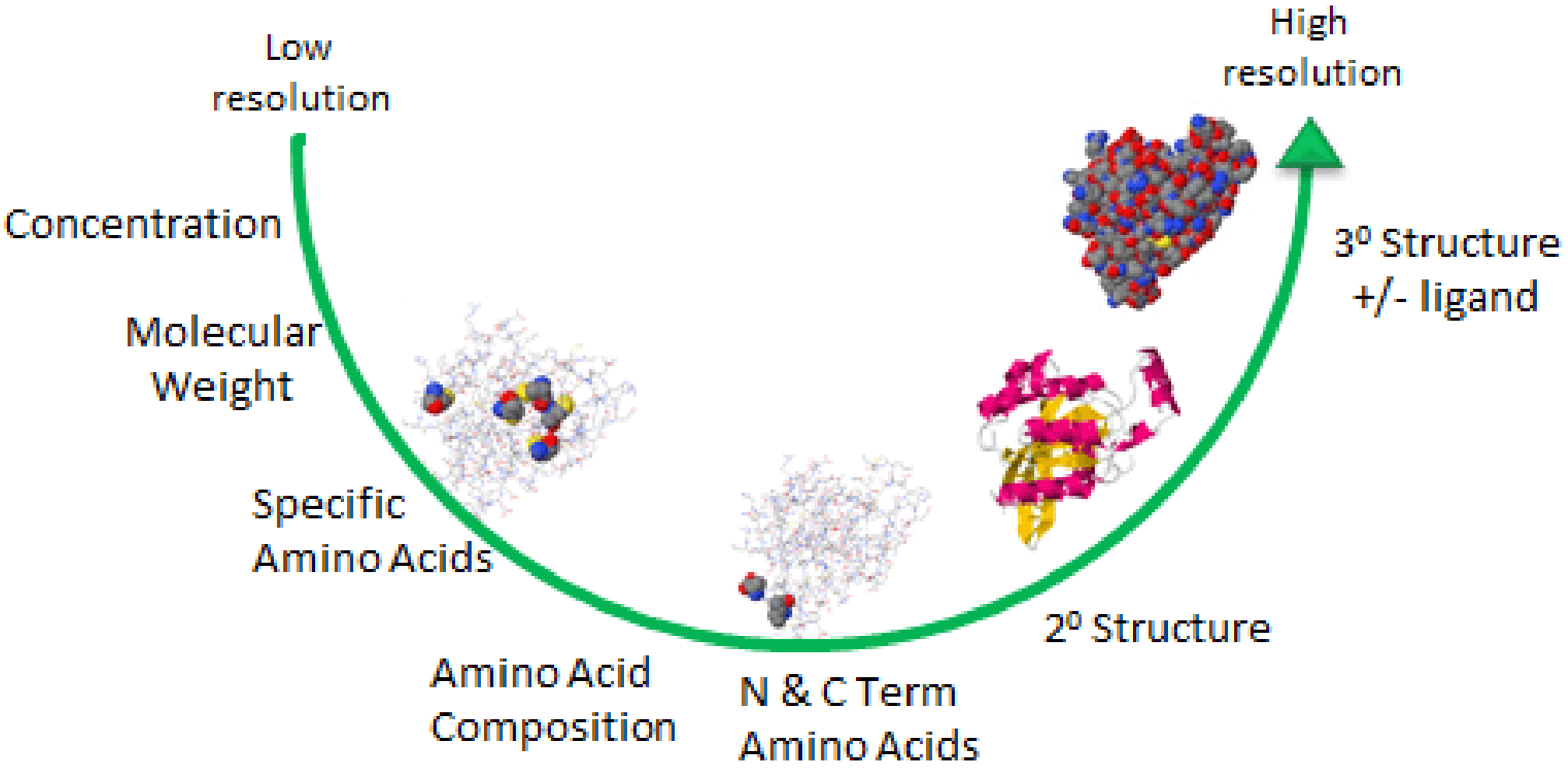
- For industrial application an enzyme, should possess some characteristics in addition to those of enzymes in the cells. These characteristics can be :
  - Enzyme should be robust with long life.
  - Enzyme should be able to use the substrate supplied in the industry even it differs from that in the cell.
  - Enzyme should be able to work under conditions, e.g., extreme of pH, temperature and concentration of the industry even if they differ from those in the cell.
  - The enzyme should be engineered to meet the altered needs. Therefore, efforts have been made to alter the properties of enzymes.
  - These are some character that one might have to change in a predictable manner in protein engineering or enzyme engineering to get the desired function.



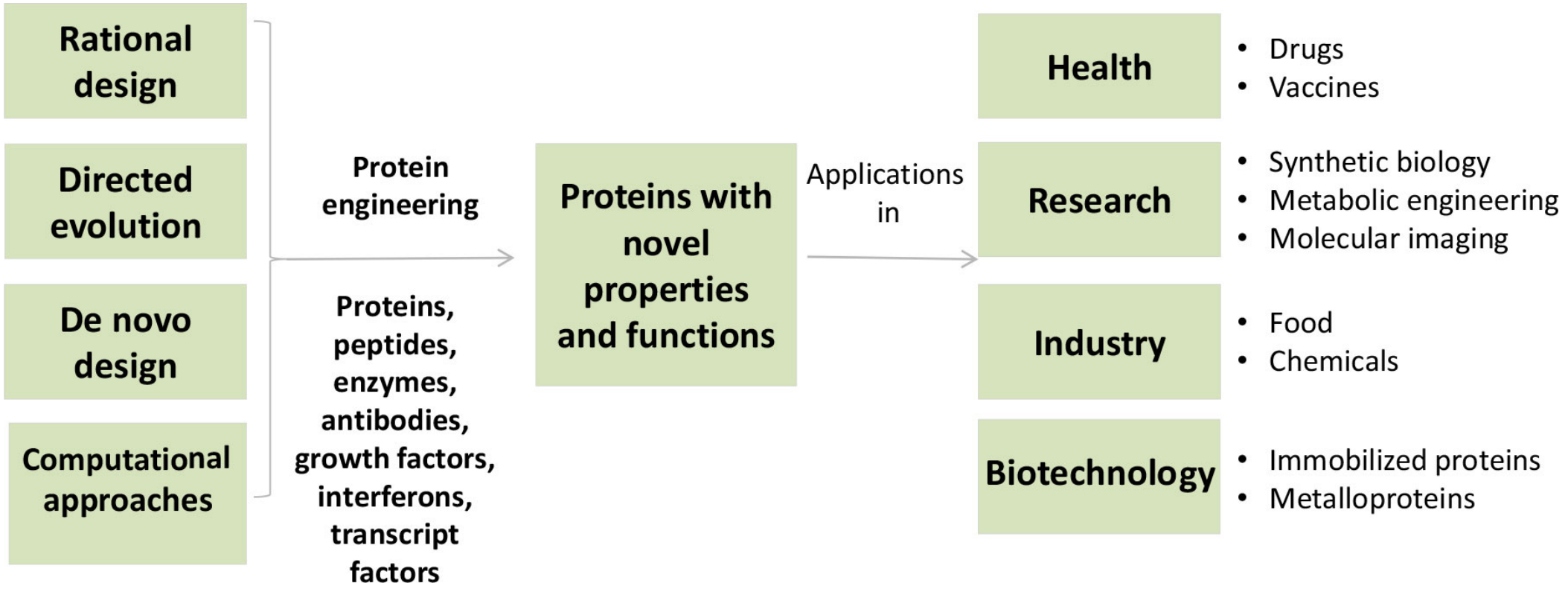
Rationale of protein engineering

- Elimination of allosteric Regulations.
- Improved kinetic properties.
- To produce biological compounds(include synthetic peptide, storage protein, and synthetic drugs) superior to natural one.
- Enhanced substrate and reaction specificity
- Increased thermostability.
- Alteration in optimum pH
- Solubility for use in organic solvents.
- Increased / decreased Optimal temperature
- To speed up the process(rate of reaction)
- Increase Protein/ Enzymes' Shelf Life
- To get high quality of Product Suitability face

Understanding the Properties of Proteins



Protein engineering



Basic assumption for protein engineering

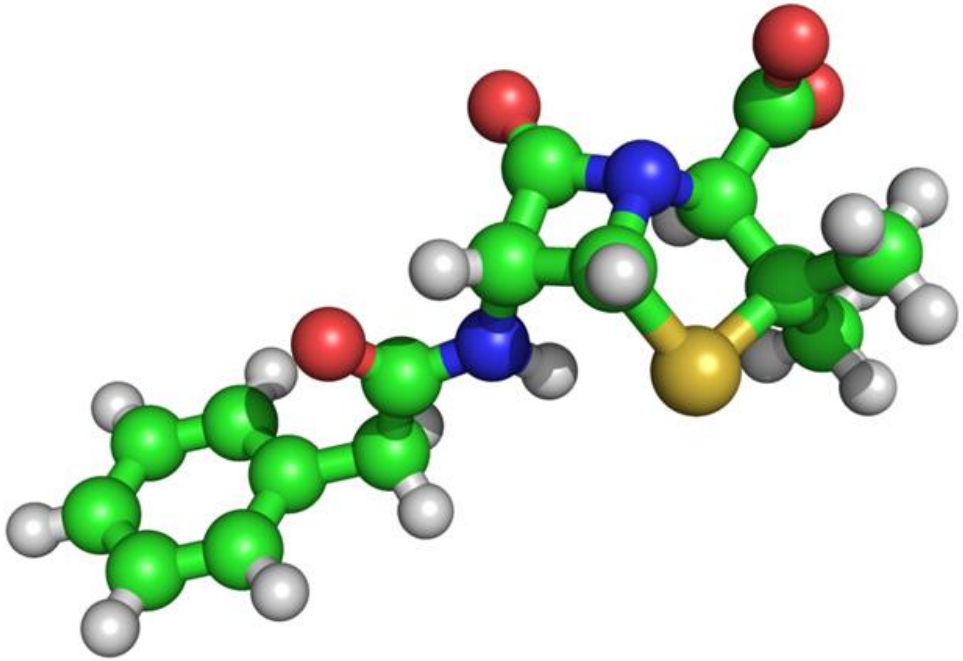
- While doing protein engineering should recognize the following properties of enzymes,
    - Many amino acid substitution, deletions or additions lead to no changes in enzyme activity so that they are silent mutator.
    - Protein have limited number of basic structures and only minor changes are superimposed on them leading to variation.
    - Similar patterns of chain folding, and domain structure can arise from different amino acid sequences with little or no homology.

Steps involved in protein engineering

- A study of three-dimensional structure of protein :

- A study of three-dimensional structure is the preliminary steps of protein engineering. And a 3d structure of protein is produced from the data generated from X-ray crystallography and NMR process by protein modeling.

- The three-dimensional structure of penicillin.



The three-dimensional structure of penicillin

Methods for protein engineering

A variety of methods are used in protein engineering such as mutagenesis, selection and recombinant DNA technology.

Mutagenesis: Mutagenesis and selection can be effectively utilized for improving a specific property of an enzyme.

Application of protein engineering

FOOD INDUSTRY

- The **wheat gluten proteins** and their heterologous expression and protein engineering has been studied using a variety of expression systems, such as E.coli, yeasts or cultured insect cells.
- Wildtype and mutant wheat gluten proteins were produced to compare them to each other for protein structure-function studies because of their availability, rapid and easy use, as well as high expression levels.
- Food-processing enzymes, such as **proteases, amylases and lipases**, the properties of which are improved using recombinant DNA technology and protein engineering.
- The deletion of native genes encoding extracellular proteases, for example, increased enzyme production yields of microbial hosts.
- Proteases are used in several applications of food industry regarding low allergenic infant formulas, milk clotting and flavors.

Application of protein engineering

FOOD INDUSTRY

- They are also important for detergent industry for removing protein stains. The improvement of proteases for industry to have, for example, high activity at alkaline pH and low temperatures, or improved stability at high temperatures is a challenge for protein engineering.
- Microbial protease production is industrially suitable because of low costs, high production yields , and easy genetic manipulation.
- Conversion of starch to bioethanol or to functional ingredients like fructose, wine, glucose and trehalose, requires microbial fermentation in the presence of biocatalysts such as amylases to liquefy and saccharify starch.

Application of protein engineering

- ENVIROMENTAL APPLICATIONS
- Genetic methods and strategies for designing microorganisms to eliminate environmental pollutants such as organic pollutants, phenols, azo dyes, organophosphorus pesticides and polycyclic aromatic hydrocarbons can be detoxified using enzymatic oxidation.
- Petroleum biorefining is also an important environmental application area, where new biocatalysts are required.
- Protein engineering, isolation and study of new extremophilic microorganisms, genetic engineering developments are all promising advances to develop new biocatalysts for petroleum refining. Petroleum biorefining applications such as fuel biodesulfurization, denitrogenation of fuels, heavy metal removal, depolymerisation.

Application of protein engineering

MEDICAL APPLICATION

- The use of protein engineering for cancer treatment studies is a major area of interest.
- Pre-targeted radioimmunotherapy and the long-circulating antibody.
- Advances in protein engineering and recombinant DNA technology were expected to increase the use of pre-targeted radioimmunotherapy.
- The use of novel antibodies as anticancer agents is also an important field of application, where the ability of antibodies to select antigens specifically and with high affinity is exploited, and protein engineering methods are used to modify antibodies to target cancer cells for clinical applications.
- Recently, the term “modular protein engineering” has been introduced for emerging cancer therapies.
- Additionally, multifunctional and smart drug vehicles can be produced at the nanoscale, by protein engineering. These strategies could be combined to identify and select targets for protein-based drug delivery.

Application of protein engineering

- MEDICAL APPLICATION
  - Pharmacokinetic properties of antibodies have been improved by protein engineering and antibody variants of different size and antigen binding sites have been produced for the ultimate use as imaging probes specific to target tissues.
  - Molecular imaging tools based on antibodies will find more applications in the future regarding diagnosis and treatment of cancer and other complex diseases.

Application of protein engineering

BIOPOLYMER PRODUCTION

- Protein engineering applications for biopolymer production are also promising.
- Protein engineering and macromolecular self assembly are utilized to produce peptide-based biomaterials, such as elastin-like polypeptides, silk- like polymers.
- The ability of protein engineering to create and improve protein domains can be utilized for producing new biomaterials for medical and engineering applications.

Application of protein engineering

NANOBIOTECHNOLOGY

- Nanotechnology application is the use of amyloid fibrils as structural templates for nanowire construction. This application is based on the fact that some proteins form well-ordered fibrillar aggregates that are called amyloid fibrils.
- The self association of well-ordered growth fibrils through noncovalent bonds under controlled conditions was suggested to have a high potential to be used for nanobiotechnology. The use of amyloid fibrils as structural templates for nanowire construction.
- Cytochrome P450 monooxygenases and protein engineering of cytochrome P450 biocatalysts for medical, biotechnological and bioremediation applications.

Thank you